Relative age of Otto stock and Matachewan dykes from paleomagnetism and implications for the Precambrian polar wander path[1,2]

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North-trending diabase dykes within and near the Otto syenite stock of the Kirkland Lake area of the Superior Province carry a stable paleomagnetic remanence identical to that of 2454 ± 2 Ma Matachewan dykes far from the stock. A positive baked contact test for the remanence of a dyke within the stock demonstrates that the remanence is primary. Thus, (1) these dykes belong to the Matachewan swarm, (2) the Otto stock has not magnetically overprinted Matachewan dykes in its contact aureole as previously reported, and (3) the stock must be older than 2454 Ma.

The magnetic remanence associated with the Otto stock, previously thought to be primary, must now be considered of uncertain age. Nevertheless, the positive baked contact test for the remanence of the Matachewan dyke within the stock demonstrates that the remanence associated with the stock predates intrusion of the Matachewan swarm.

The conventional polar wander path for Precambrian North America decreases in age from the Matachewan paleopole to the Otto stock paleopole. Since this study demonstrates that the Otto stock pole is older than the Matachewan pole, there is a serious problem with the conventional path.

Les dykes de diabase de direction nord à l'intérieur et au voisinage du petit massif de la syénite d'Otto dans la région de Kirkland Lake, province du Supérieur, portent une rémanence paléomagnétique stable identique à celle des dykes de Matachewan (2454 ± 2 Ma) éloignés de ce petit massif. L'application d'un test du contact pour la rémanence d'un dyke localisé à l'intérieur du petit massif prouve qu'elle est primaire. Donc, (1) ces dykes appartiennent à l'essaim de Matachewan, (2) le petit massif d'Otto n'a pas créé de surimpression magnétique sur les dykes de Matachewan dans l'auréole de contact, contrairement à ce fut rapporté jadis et (3) le petit massif doit être plus vieux que 2454 Ma.

La rémanence magnétique associée au petit massif d'Otto, interprétée jadis comme primaire, doit maintenant être considérée d'âge incertain. Néanmoins, le test de contact appliqué pour la rémanence du dyke de Matachewan à l'intérieur du petit massif démontre que la rémanence associée au petit massif fut acquise antérieurement à l'intrusion de l'essaim de Matachewan.

La courbe conventionnelle de dérive des pôles pour le Précambrien de l'Amérique du Nord décroît en âge à partir du paléopôle de Matachewan au pôle du petit massif d'Otto. Vu que cette étude montre que le pôle du petit massif d'Otto est plus vieux que le pôle de Matachewan, alors un problème sérieux existe au sujet de cette courbe conventionnelle de dérive des pôles.

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Introduction

This study reexamines the positive baked contact test reported by Pullaiah and Irving (1975) for the remanent magnetization associated with the Otto syenite stock of the Abitibi Belt in the Superior Province. The original test was based on the apparent overprinting of a single diabase dyke in the contact aureole of the stock.

Pullaiah and Irving (1975) established a direction of remanent magnetization for rocks associated with the Otto stock. At that time the stock was thought to be 2114 ± 80 Ma on the basis of Rb/Sr dating (Bell and Blenkinsop 1976). Although the syenite of the stock is weakly and unstably magnetized, late-stage lamprophyre dykes within the stock and Archean metavolcanics in the contact aureole carry a remanence that is steeply down to the north-northwest.

The positive baked contact test of Pullaiah and Irving (1975) indicated that the steep down magnetization associated with the stock is primary. A diabase dyke in the contact aureole, thought to belong to the Matachewan swarm, was shown to carry a steep down magnetization, unlike the shallow south-southwest remanence typical of Matachewan dykes from this portion of the Abitibi Belt (e.g., Irving and Naldrett 1977). Therefore, the dyke in the contact aureole was assumed to be magnetically overprinted at the time that the stock intruded.

In addition to the steep down magnetization, three of the lamprophyre dykes within the Otto stock have a soft component, somewhat similar in direction to the typical shallow south-southwest Matachewan remanence. Pullaiah and Irving (1975) interpreted this component as a later secondary overprint, although they recognized the difficulty posed by the presence of a Matachewan-like component within the stock, which was thought to be younger than the Matachewan dyke swarm.

In this study, north-trending diabase dykes were sampled over a broad area in the vicinity of the stock. During the course of the fieldwork, three north-trending diabase dykes were located within the stock itself. They were sampled even though one had previously been mapped as "Keweenawan or Nipissing" (Ontario Department of Mines Map 2239,1972). At one of these sites the diabase dyke cuts a lamprophyre dyke, allowing a baked contact test to be carried out.

Geological setting

A major swarm of north- to northwest-trending diabase dykes intrudes Archean rocks in the central Superior and Southern provinces (Fig. 1). The northwest-trending dykes close to Lake

[1]Geological Survey of Canada Contribution 40889.
[2]Publication 17 of International Geological Correlation Program (IGCP) Project 257 on Precambrian Dyke Swarms.

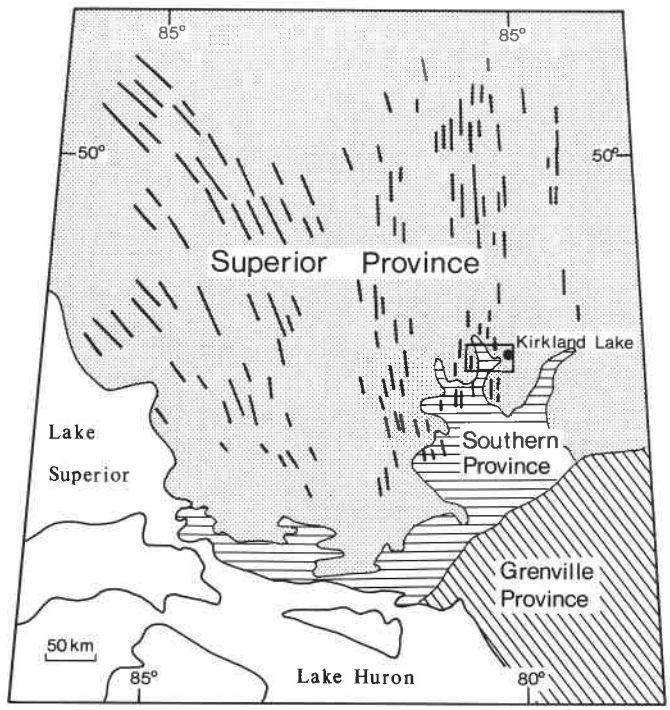


FIG. 1. The Matachewan–Hearst dyke swarm of the central Superior and Southern provinces (after Fahrig and West 1986). The sampling area in the box is enlarged in Fig. 2.

Superior are usually referred to as Hearst dykes, while north-trending dykes farther east are termed Matachewan dykes. The Matachewan portion of the swarm has been correlated on the basis of paleomagnetic, petrographic and aeromagnetic data with the Hearst portion of the swarm (Ernst and Halls 1984; Watson 1980). A U/Pb zircon age of 2454 ± 2 Ma has recently been obtained for a Hearst dyke (Heaman 1988). Matachewan dykes are quartz tholeiites and often contain large feldspar phenocrysts (Card 1981).

The study area is located west of Kirkland Lake, on the boundary between the Superior and Southern provinces (Fig. 1). Here the Otto syenite stock intrudes Archean metavolcanics of the Abitibi Belt (Fig. 2). North-trending Matachewan dykes occur frequently in the country rock. The geology of the area has been described in detail by Lovell (1967, 1972).

The Otto stock is about 11 km in diameter and ranges in composition from quartz syenite to nepheline syenite. It contains many Archean metavolcanic and metasedimentary inclusions. Lamprophyre dykes, which are thought to be due to more pervasive contamination of the syenite by country rock (Lovell 1972), are particularly well exposed along Highway 112 in the eastern portion of the body.

Bell and Blenkinsop (1976) reported a whole-rock Rb/Sr age of 2114 ± 80 Ma (recalculated with decay constant of Steiger and Jäger 1977) for the Otto stock. This age was used by Pullaiah and Irving (1975) and subsequently by others (e.g., Irving and McGlynn 1981) in discussions of the Otto stock paleopole. Earlier studies yielded widely divergent ages. Purdy and York (1968) obtained a whole-rock Rb/Sr isochron age of 1693 ± 50 Ma (recalculated), while Aldrich and Wetherill (1960) determined a K/Ar age of 2500 ± 130 Ma and a Rb/Sr age of 2418 ± 120 Ma (recalculated) on biotite from a biotite-rich inclusion in the stock. While the present study was in progress we learned that a precise U/Pb zircon age of 2680 ± 1 Ma has recently been obtained for the Otto stock (Corfu *et al.* 1989).

Archean metavolcanics are regionally metamorphosed to chlorite–albite grade (Lovell 1972). Adjacent to the stock, however, there is a contact aureole of garnet–epidote amphibolite grade with a width of up to 0.8 km. Here, the metavolcanics are steeply folded.

Sampling and laboratory procedure

Samples were collected during the 1988 field season from north-trending diabase dykes along two profiles (Fig. 2), one approaching the Otto stock from the west, the other from the north. The east–west profile consists of 13 sampling sites taken at distances between ~500 m and 35 km from the stock. Four sites were sampled along the north–south profile at distances of ~300 m to 21 km from the stock. Most dykes were located on Ontario Department of Mines Maps 2060, 2078, 2110, 2115, and 2239.

Three north-trending diabase dykes were also collected within the Otto stock. Two of these dykes (sites 17 and 18) do not appear on Ontario Department of Mines Map 2239 of the stock, while the third (site 5) is identified as a "Keweenawan or Nipissing" dyke. At site 18, a vertical north-trending diabase dyke cuts a northwest–trending lamprophyre dyke that dips ~55° southwest. Drill cores were collected from the diabase and in the lamprophyre at increasing distances from the diabase contact to carry out a baked contact test.

At each site, five or more block samples or drill cores were oriented with sun and magnetic compasses. About three cylindrical specimens (2.2 cm length and 2.5 cm diameter) were cut from each sample or core.

Remanent magnetization was measured using a Schonstedt DSM1 commercial spinner magnetometer. Each specimen was cleaned using stepwise alternating-field (AF), thermal, or two-stage AF–thermal demagnetization, depending on which procedure was most efficient in eliminating unstable components of magnetization. AF cleaning was carried out at peak fields of 2.5, 5.0, 10, 15, 20, 25, 30, 35, 40, 50, 60, 70, 80, 90, and 100 mT using a Schonstedt GSDl demagnetizer. Thermal steps of 200, 300, 400, 450, 500, 530, 550, (555), 560, (565), 570, 580, 590°C were employed in a Schonstedt furnace modified to control temperatures to ±5°C. In some sites a stable magnetization was most easily isolated by AF demagnetizing between 5 and 30 mT before the thermal procedure was begun. All measurements were carried out in a shielded room with walls consisting of two layers of electrical-grade steel and a magnetic field less than 1000 nT. Helmholtz coils reduced the field to less than 50 nT over a small working area.

Horizontal and vertical projections of demagnetization data for individual specimens were plotted on orthogonal-component plots. End points were then established from straight-line segments using a three-dimensional least square fitting program (Kirschvink 1980).

Paleomagnetic results

Seventeen of the 20 diabase sampling sites carry stable directions of remanent magnetization. Two of the remaining sites (sites 13 and 15) are unstably magnetized, while the third (site 25) carries a semistable anomalous direction. The latter site is located far from the Otto stock (Fig. 2).

At most sites a soft magnetization component is removed in AF's less than 15 mT. This component often has a steep down direction that is rather scattered. It is interpreted as a viscous

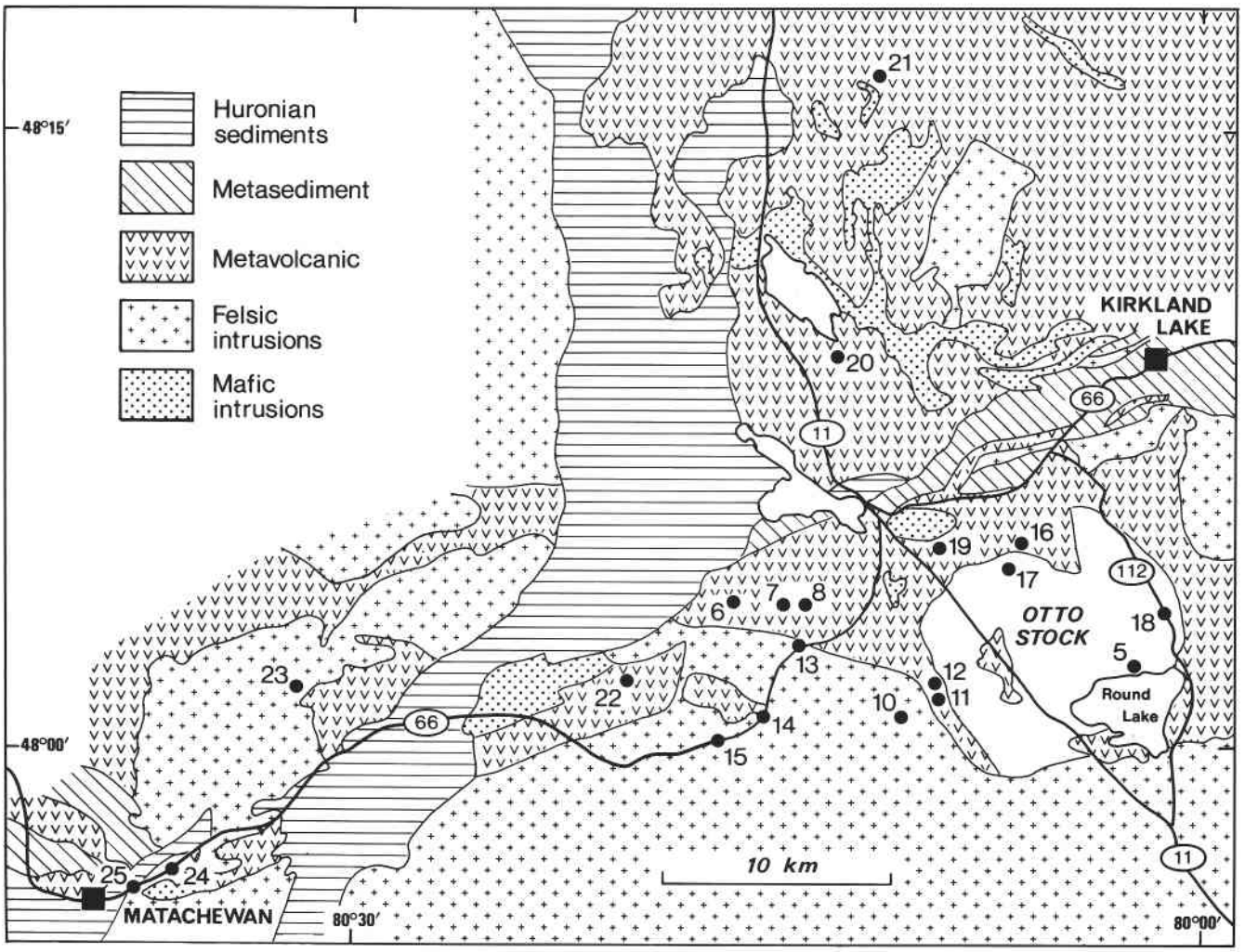


FIG. 2. Geology and sampling sites in the Matachewan – Kirkland Lake area. Geology after Ontario Department of Mines Map 2205. All sampling sites are in north-trending diabase dykes. A baked contact test between a north-trending diabase dyke and an older lamprophyre dyke that intrudes the Otto syenite stock was conducted at site 18.

magnetization acquired in the present Earth's magnetic field. Occasionally the viscous component persists to somewhat higher AF's and may significantly overlap the coercivity range of the harder component described below.

After removal of the viscous component, a hard magnetization with a shallow south-southwest direction is isolated in all 17 stable sites. At some sites the hard magnetization is most clearly resolved using thermal demagnetization or two-stage AF–thermal cleaning.

Typical examples of demagnetization data are shown for sampling sites located far from the Otto stock, adjacent to the stock, and within the stock in Figs. 3, 4, and 5 respectively. No difference in the direction of the hard component of magnetization could be detected with distance from the stock. Indeed, the hard magnetization is indistinguishable from that observed in Matachewan dykes in other areas of the Abitibi Belt (Irving and Naldrett 1977; Schutts and Dunlop 1981), or elsewhere in the Superior Province (e.g., Halls and Shaw 1988).

At site 18, within the Otto stock, a diabase dyke cuts an older lamprophyre dyke (Fig. 6) so that a baked contact test is possible. A shallow south-southwest magnetization is observed in both the diabase (cores 1–5; Figs. 6, 7*a*) and adjacent baked lamprophyre (core 6; Figs. 6, 7*b*). On the other hand, unbaked lamprophyre further from the diabase contact (cores 7–9; Figs. 6, 7*c*) carries a steep north-northwest direction. This direction is similar to that obtained by Pullaiah and Irving (1975) in late-stage lamprophyre dykes within the Otto stock and in metavolcanics in the contact aureole, and was interpreted as representing the time of intrusion of the stock.

The positive baked contact test at site 18 demonstrates that the shallow south-southwest (Matachewan) remanence of the diabase dyke is primary. It also indicates that the steep north-northwest remanence of the unbaked lamprophyre dyke predates intrusion of the diabase.

Analysis

The direction of stable magnetization in individual specimens was determined from straight-line segments on orthogonal-component plots using a three-dimensional least squares fitting program (Kirschvink 1980). Two (or occasionally three) specimen directions were averaged to yield a mean sample direction. Sample directions were then averaged to give mean site directions.

The 17 site means representing the shallow south-southwest Matachewan direction of magnetization are listed in Table 1 and plotted in Fig. 8*a*. There is no systematic difference among remanence directions obtained far from the Otto stock, close to

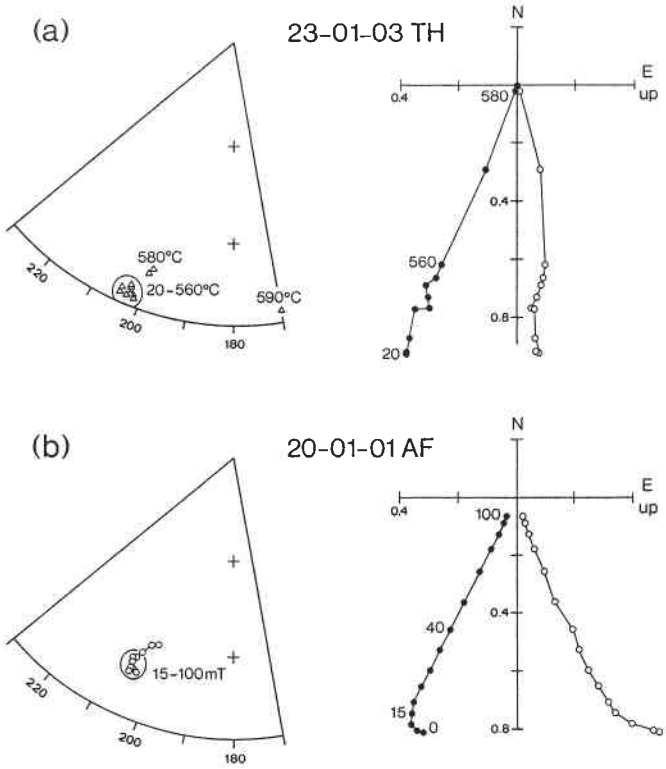


FIG. 3. Characteristic paleomagnetic results for dykes far from the Otto stock: (*a*) thermal portion of AF–thermal demagnetization for specimen 01-03 at site 23, collected 27 km from the stock; (*b*) AF demagnetization of specimen 01-01 at site 20, collected 10 km from the stock. Directions are plotted on an equal-area net with closed (open) symbols indicating downward (upward) directions and circles (triangles) indicating AF (thermal) demagnetization. Orthogonal-component plots show the data projected onto the horizontal plane (closed symbols) and the north-up vertical plane (open symbols). Data are plotted relative to initial intensity of natural remanent magnetization. A straight-line segment on the orthogonal-component plot represents removel of a single component.

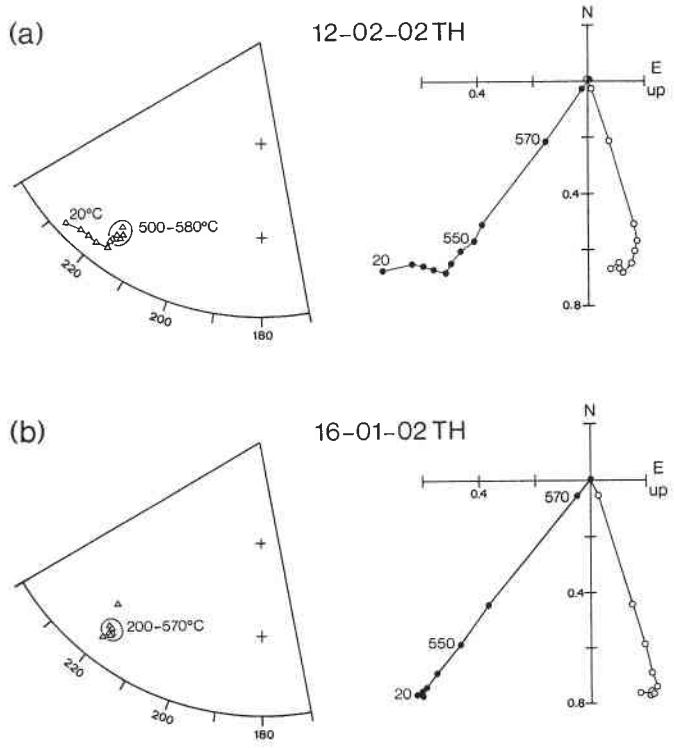


FIG. 4. Characteristic paleomagnetic results for dykes close to the Otto stock. Thermal portion of AF–thermal demagnetization of (*a*) specimen 02-02 at site 12, collected about 500 m from the Otto stock and (*b*) specimen 01-02 at site 16, collected about 300 m from the stock. Symbols are the same as in Fig. 3.

the stock, or within the stock. Sixteen sites were used to calculate a mean direction of $D = 209.5°$, $I = -14.8°$ ($k = 39$, $\alpha_{95} = 6.0°$), shown as a star in Fig. 8*a*. Site 11 was not included because of its high error value ($\alpha_{95} = 41°$).

For comparison, mean magnetization directions reported in two studies of Matachewan dykes from the Abitibi Belt (Irving and Naldrett 1977; Schutts and Dunlop 1981) are shown in Fig. 8*b*. Similarly, the Matachewan-like overprint reported by Pullaiah and Irving (1975) at three lamprophyre dyke sites within the Otto stock is also illustrated in Fig. 8*b*.

The steep north-northwest remanence direction obtained for unbaked lamprophyre samples at site 18 of the present study is plotted in Fig. 8*a*. It is in excellent agreement with that reported by Pullaiah and Irving for other lamprophyre dykes within the Otto stock and metavolcanics in the contact aureole (Fig. 8*b*).

Discussion

Pullaiah and Irving (1975) reported a steep north-northwest remanence ($D = 330°$, $I = 71°$, $\alpha_{95} = 3°$) in lamprophyre dykes and metamorphosed volcanics associated with the Otto stock. In addition, they obtained a steep remanence ($D = 273°$, $I = 59°$) at a single dyke site in the contact aureole of the Otto stock. On

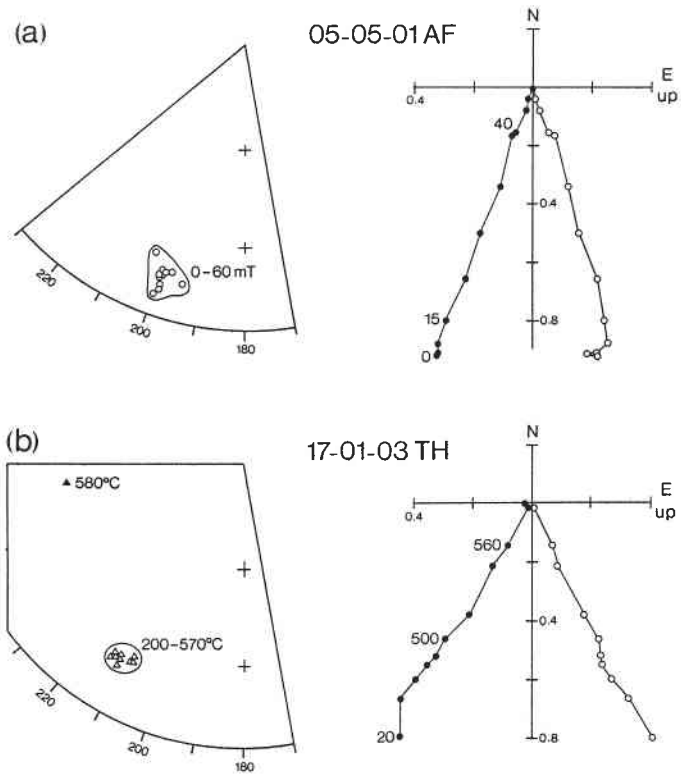


FIG. 5. Characteristic paleomagnetic results for north-trending diabase dykes within the Otto stock: (*a*) AF demagnetization of specimen 05-01 at site 05; (b) thermal demagnetization of specimen 01-03 at site 17. Symbols are the same as in Fig.3.

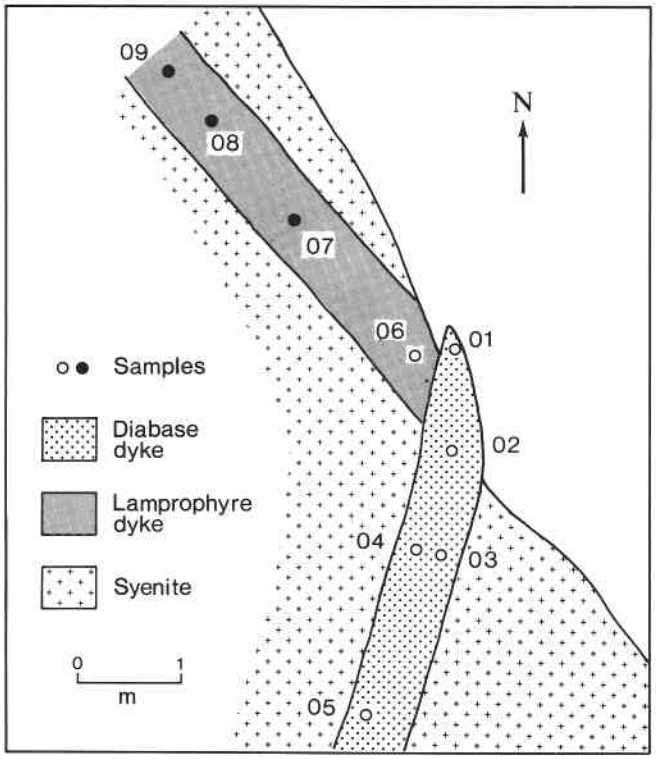


FIG. 6. Geology for baked contact test at site 18 showing plan view of road cut on southwest side of Highway 112. The older lamprophyre dyke dips 55° southwest and is cut by the younger vertical diabase dyke. Drill cores which carry a Matachewan direction of magnetization are shown as open circles, those with a steep north-northwest direction are shown as closed circles.

this result they based a positive baked contact test for the Otto stock remanence. However, the remanence of this site, which is distinct from the grouping of remanence directions associated with the stock, could also be explained as a ~2150 Ma Preissac or a 1140 Ma Abitibi dyke remanence (Larochelle 1966).

The present study demonstrates that Matachewan dykes within the contact aureole of the Otto stock have retained a stable primary magnetization with a shallow south-southwest direction. Further, it is shown that north-trending dykes within the stock belong to the Matachewan dyke swarm, since they carry a primary Matachewan remanence (see baked contact test at site 18). Both of these observations indicate that the baked contact test of Pullaiah and Irving (1975) is invalid and that the Otto stock must be older than 2454 Ma, the age of Matachewan dykes (Heaman 1988). The latter conclusion has recently been confirmed in an independent U/Pb zircon study of the Otto stock in which an age of 2680 ± 1 Ma has been obtained (Corfu *et al.* 1989).

If the baked contact test for the remanence associated with the Otto stock is invalid, then the primary or secondary nature of this remanence is uncertain. Nevertheless, the positive baked contact test between the Matachewan dyke and the lamprophyre dyke that cuts the Otto stock at site 18 demonstrates that the Otto stock pole is older than the Matachewan pole.

This conclusion has important implications for the late Archean – Early Proterozoic apparent polar wander path for North America. The conventional path (Fig. 9) follows that of Irving (1979) from the Matachewan paleopoles (MD1, MD2, and MD3) to the paleopole associated with the Otto stock (OS). More recent versions of the path (e.g., Irving and McGlynn 1981; Dunlop and Schutts 1983; Piper 1987) are generally similar to that of Irving (1979). Results of the present study

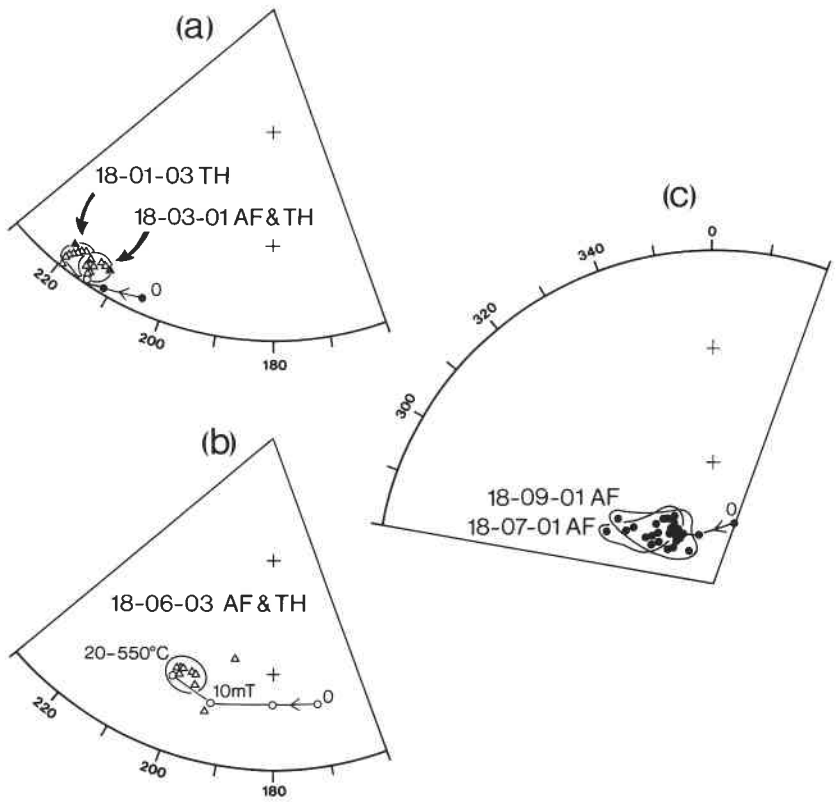


FIG. 7. Paleomagnetic directions for the baked contact test at site 18: (*a*) diabase dyke; (*b*) baked lamprophyre dyke; (*c*) unbaked lamprophyre dyke. Triangles indicate thermal demagnetization, circles indicate AF demagnetization.

TABLE 1. Summary of paleomagnetic results

Site	Procedure	N	n	D (°)	I (°)	k	α_{95} (°)
05	AF	5	5	197.4	−20.4	346	4.1
06	AF–TH	5	3	216.1	−3.1	56	16.6
07	AF	5	5	206.8	−29.8	245	4.8
08	AF	5	5	219.3	−27.7	170	5.8
10	AF	6	5	200.0	−25.9	97	7.8
11	AF	5	3	205.5	−14.4	10	41.1
12	AF–TH	5	5	215.5	−12.2	35	13.2
14	AF–TH	5	5	202.9	−17.4	321	4.2
16	AF–TH	5	5	217.7	−12.9	115	7.1
17	TH	5	4	215.4	−20.1	43	14.1
18[a]	AF–TH	6	6	211.3	−12.8	85	7.2
19	AF	5	4	217.5	−23.8	80	10.3
20	AF	5	5	213.8	−13.4	20	17.7
21	AF	5	5	191.5	−14.8	25	15.4
22	AF	5	4	200.0	−4.2	55	12.4
23	AF–TH	5	5	211.1	−2.1	19	17.9
24	AF	5	4	215.4	6.9	49	13.2
Mean[b]		20	16	209.5	−14.8	39	6.0
Paleopole		58.4°E		42.3°N	dm = 6.2°	dp = 3.2°	

NOTES: Procedure is demagnetization method used (AF, alternating field; TH, thermal); N is number of samples (sites) measured; n is number of samples (sites) used; D is magnetic declination; I is magnetic inclination; k is precision parameter; α_{95} is radius of circle of 95% confidence; dm and dp are the semiaxes of the oval of 95% confidence.

[a]Site 18 mean based on five diabase and one overprinted lamprophyre sample. Three other lamprophyre samples at site 18 that have not been overprinted by the diabase yield a mean direction of D = 338.8°, I = 75.2°, k = 544, and α_{95} = 5.2°, and a corresponding paleopole at 112.5°W, 71.7°N, dm = 9.8°, dp = 6.5°.

[b]Mean direction is based on sites 05–08, 10, 12, 14, and 16–24. Site 11 is excluded because of high α_{95}.

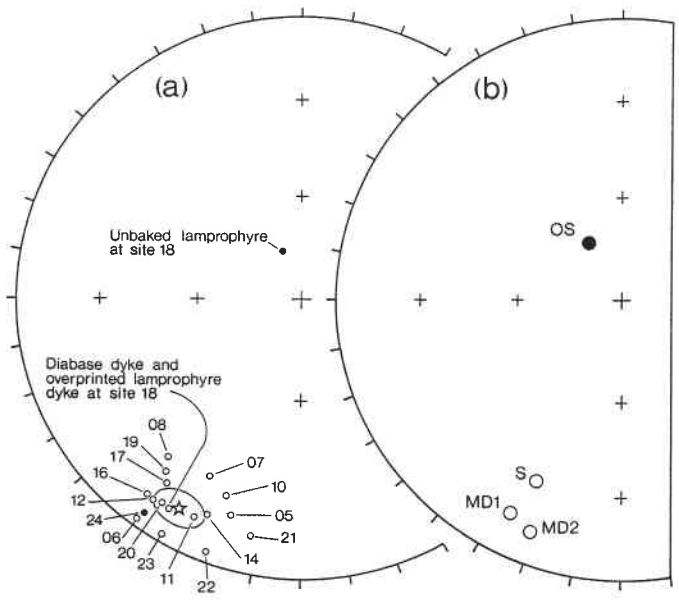


FIG. 8. (*a*) Mean directions of magnetization at individual sampling sites for present study. The mean of site directions (excluding the unbaked lamprophyre at site 18) is shown as a star with its circle of 95% confidence. (*b*) For comparison, mean directions obtained in two studies of Matachewan dykes in the Abitibi Belt (MD1, Irving and Naldrett 1977 and MD2, Schutts and Dunlop 1981) are shown, as is the mean direction associated with the Otto stock (OS) as observed in lamprophyre dyke within the stock and in metavolcanics in the contact aureole (Pullaiah and Irving 1975). A soft overprint observed by Pullaiah and Irving (1975) in three lamprophyre dykes (S) is also plotted.

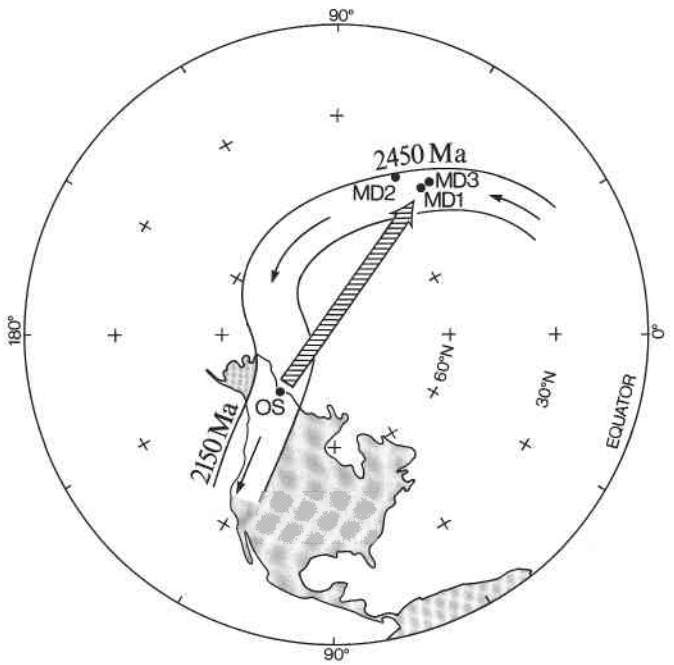


FIG. 9. Matachewan dyke paleopoles for the Abitibi Belt (MD1, Irving and Naldrett 1977; MD2, Schutts and Dunlop 1981; MD3, present study) are compared with the pole associated with the Otto stock (OS, Pullaiah and Irving 1975). The conventional apparent polar wander path from about 2500 to 2150 Ma (Irving 1979) is superimposed with dating of the path modified to reflect the revised age of 2454 Ma for Matachewan dykes (Heaman 1988). The present study demonstrates that OS is older than MD (see hachured arrow), the reverse of what might be expected from the conventional path.

cannot easily be reconciled with the conventional path, since the paleopole for the Otto stock is clearly older than the Matachewan pole (see hachured arrow of Fig. 9).

Many of the other paleopoles on which the 1979 path or later versions are based have since been reinterpreted, often in light of more precise radiometric age determinations. For example, the paleopole for the Shelley Lake Granite was originally interpreted as a primary paleopole about 2700 Ma in age (Dunlop 1979). It has since been shown to be a secondary overprint at about 1200 Ma (Dunlop 1984). Likewise, the paleopole for the Thessalon volcanics, considered by Symons and O'Leary (1978) as a primary pole at about 2375 Ma, has subsequently been reinterpreted (Stupavsky and Symons 1983) as a secondary pole.

Indeed, few paleomagnetic tests (baked contact tests, fold tests, or conglomerate tests) have been conducted to establish the primary or secondary nature of the remanence of late Archean – Early Proterozoic units. Therefore, the age of most remanences on the late Archean – Early Proterozoic polar wander path for North America is poorly constrained.

Some paleopoles used in the conventional path are based on very scanty data. For example, the pole for the Coleman Member of the Gowganda Formation is derived from only a single sampling site (Roy and Lapointe 1976). Other paleopoles, such as that for the Algoma intrusions (Londry and Symons 1976), have only been reported in abstract form and, hence, are difficult to evaluate.

It is also not clear to what extent the conventional polar wander path for the late Archean – Early Proterozoic actually represents the entire Canadian Shield. Most of the data come from the southern Superior Province and the Southern Province. It is perhaps best, therefore, to refer to a south Superior – Southern apparent polar wander path, until such time as well-dated paleopoles from other parts of the Shield become available.

In any case, the conventional path cannot, in its current form, explain the Matachewan – Otto stock results of the present study. Until the paleomagnetic data base for the late Archean – Early Proterozoic is improved, it will be difficult to determine whether the conventional path can be modified with the inclusion of another loop to accommodate an Otto stock paleopole that is older than that for Matachewan dykes or whether a completely new path should be drawn. It might be noted, however, that a new path for this period has recently been proposed by Costanzo-Alvarez and Dunlop (1989) on the basis of paleomagnetic data from the Red Lake greenstone belt of northwestern Ontario.

Conclusions

Several important conclusions can be derived from the paleomagnetism of north-trending diabase dykes in and near the Otto syenite stock:

(1) North-trending diabase dykes that cut the Otto stock and that previously were mapped as "Nipissing or Keweenawan" are shown, from their paleomagnetic signature, to belong to the 2454 Ma Matachewan swarm.

(2) The magnetization of the Matachewan dykes of the study area is primary, as demonstrated by the positive baked contact test at site 18.

(3) The Otto stock, formerly thought to be younger than Matachewan dykes, must, in fact, be significantly older than the dykes.

(4) The magnetization of lamprophyre dykes that cut the Otto stock is pre-Matachewan, as shown by the baked contact test at site 18.

(5) Whether the magnetization of the lamprophyre dykes is primary is no longer clear, since the positive baked contact test reported by Pullaiah and Irving (1975) must now be considered invalid.

(6) The conventional apparent polar wander path for North America (e.g., Irving 1979), which decreases in age from the Matachewan paleopole to that for the Otto stock, cannot be easily reconciled with the relative ages of the magnetizations of these two units from paleomagnetism.

Acknowledgments

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